

Task 18 — Market Basket Analysis & Cross-Sell Strategy Report

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1. EXECUTIVE SUMMARY & OBJECTIVE

Objective: Perform association rule mining on basket transaction logs to identify high-co-occurrence item pairings, power personalized recommendations, and design high-margin product bundles.

Methodology: Applied the **Apriori algorithm** (Min Support = 0.10) followed by rule filtering (Min Confidence = 0.50, Min Lift > 1.3) to avoid high-confidence/low-lift trivia.

2. TOP VETTED ASSOCIATION RULES

ANTECEDENT (IF)	CONSEQUENT (THEN)	SUPPORT	CONFIDENCE	LIFT	STRATEGIC RECOMMENDATION
Monitor	USB Cable	0.33	0.94	2.77	Hardware Bundle: Checkout Pop-up
Laptop, Mouse	Keyboard	0.31	0.71	1.56	Workstation Bundle: 10% Off Kit
Laptop	Mouse	0.44	0.97	1.49	Default Cart Recommendation
Keyboard	Mouse	0.34	0.75	1.16	Accessory Shelf Placement

3. BUSINESS INTERPRETATION & PITFALL SAFEGUARDS

- **High-Lift Prioritization:** Filtered out redundant item combinations (e.g., universal add-ons with Lift near 1.0) to prevent rule overload.
- **Bundle 1 (Display Essentials):** Monitor $\rightarrow$ USB Cable (Lift = 2.77, Confidence = 94%) represents the strongest affinity pair. Recommend auto-adding cables at checkout.
- **Bundle 2 (Workstation Setup):** Laptop + Mouse $\rightarrow$ Keyboard (Lift = 1.56, Confidence = 71%) supports creating a unified Work-From-Home kit.

4. DELIVERABLES & REPOSITORY VERIFICATION

- **Output Files:** basket_transactions.csv and market_basket_rules.csv.
- **GitHub Repository:** Live on <https://github.com/JeevanN-07/placemux.git>.